

# CT214H – Web Programming

## Practical Session 4 – MySQL & PHP

### Content

- ❖ PHP - MySQL basics: create a connection, create a database, manipulate data, execute queries, ...
- ❖ Prerequisites: XAMPP installed. Your starting point should be from the Techblog web page in the previous practical session.
- ❖ **Objective:**
  - Create database and tables
  - Connect to database
  - Query data and display on your page
- ❖ **Requirement:**
  - Step-by-step do the tasks below and observe the results

### 1. Database setup

#### a. Create database

- Open phpMyAdmin (<http://localhost/phpmyadmin>)

```
CREATE DATABASE techblog_db;  
USE techblog_db;
```

- Run this SQL query in the SQL tab

#### b. Create Articles table

This table is designed to store and manage articles. Each record represents a single article. The table uses an auto-incrementing id as the primary key to uniquely identify each article. Core content fields such as title, category, excerpt, and content store the article's headline, classification, short preview, and full text respectively.

```
CREATE TABLE articles (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  title VARCHAR(255) NOT NULL,  
  category VARCHAR(100) NOT NULL,  
  excerpt TEXT NOT NULL,  
  content TEXT NOT NULL,  
  author VARCHAR(100) NOT NULL,  
  image_url VARCHAR(255),  
  views INT DEFAULT 0,  
  likes INT DEFAULT 0,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP);
```

### 2. Insert sample data

a. Add sample data into Articles table: [sampledata.txt](#)

b. Verify data was inserted

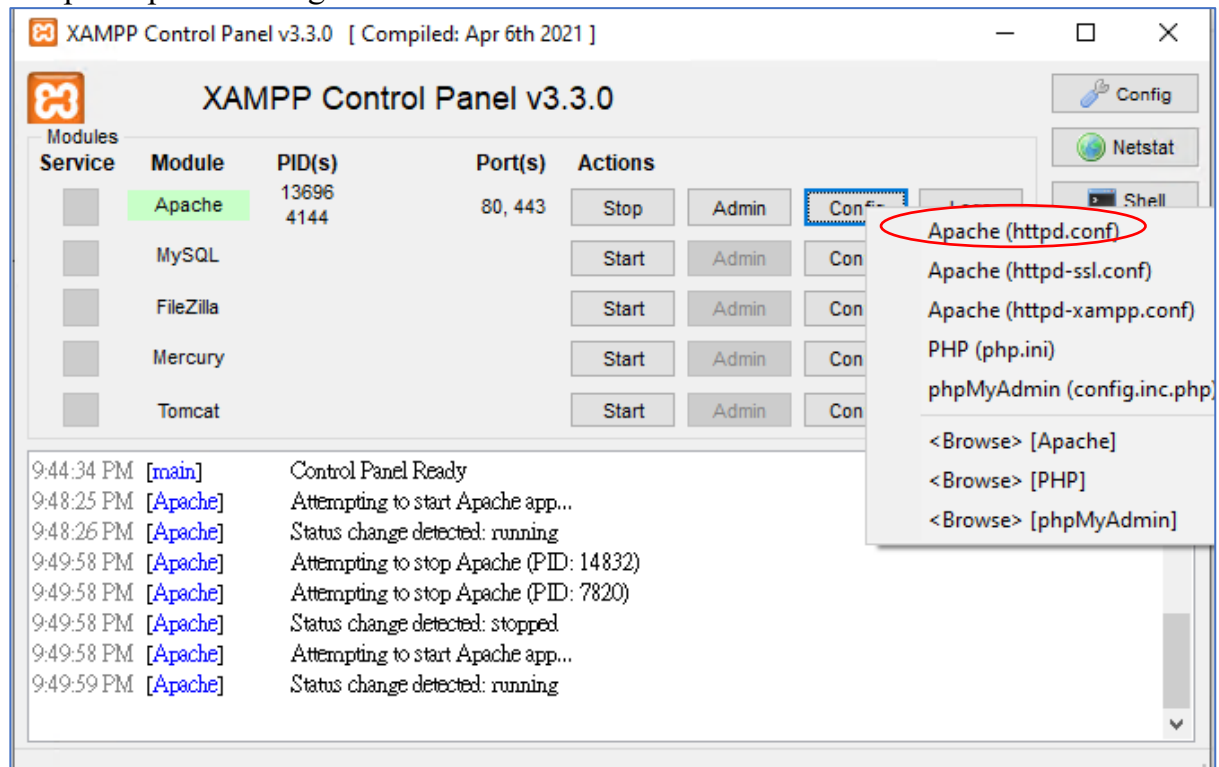
Write a SQL query to:

- Select all articles

- Select all articles in the category 'Web Development'
- Select all articles with the following information: title, author, category, created\_at
- Select all articles with zero views
- Select all articles whose title contains at least one of the following words: Cloud, Security, Web
- Select all articles whose title or excerpt contains at least one of the following words: Cloud, Security, Web

### 3. Setting the Path to the Source Code Folder

a. Open Apache configuration file



b. Add to the end of the file (with StudentNumber is your Student number):

```
Alias /myblog D:\WebProgramming\StudentNumber
<Directory D:\WebProgramming\StudentNumber>
    Options Indexes FollowSymLinks ExecCGI Includes
    AllowOverride All
    Require all granted
</Directory>
```

c. Save and restart your Apache server

After completing the above steps, you can access the files in the D:\WebProgramming\StudentNumber folder from your web browser using the following link: <http://localhost/myblog/>

### 4. PHP database connection

a. Create a php file (db-connect.php)

```
$host = 'localhost'; // Database host (usually localhost)
$username = 'root'; // MySQL username (default is 'root')
$password = ''; // MySQL password (default is empty for XAMPP)
$dbname = 'techblog_db'; // Database name
// Create connection
$conn = mysqli_connect($host, $username, $password, $dbname);
// Check connection
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}
// Success message (remove this in production)
//echo "Connected successfully!";
```

b. Test the connection

Create a test-connection.php

```
// Include database connection
include 'db-connect.php';
// Test query
$sql = "SELECT COUNT(*) as total FROM articles";
$result = mysqli_query($conn, $sql);
$row = mysqli_fetch_assoc($result);
echo "Database connected successfully!<br>";
echo "Total articles in database: " . $row['total'];
// Close connection
mysqli_close($conn);
```

## 5. Display articles from database

1. Rename your techblog.html to techblog.php (Your CSS and JS will still work!)
2. Replace static HTML articles with PHP

ORIGINAL HTML (remove this):

```
<article class="article-card">
  
  <div class="article-body">
    <span class="category">Web Development</span>
    <h3 class="article-title">Modern CSS Techniques</h3>
    <p class="article-excerpt">Learn about...</p>
    <a href="#" class="read-more">Read More →</a>
  </div>
</article>
```

Replace with php code:

```
<?php
// Include database connection
include 'db-connect.php';
// Query to get all articles
$sql = "SELECT * FROM articles";
$result = mysqli_query($conn, $sql);
// Check if we have articles
if (mysqli_num_rows($result) > 0) {
// Loop through each article
while ($article = mysqli_fetch_assoc($result)) { ?>
    <article class="article-card">
        "
            class="article-image">
        <div class="article-body">
            <span class="category"><?php echo htmlspecialchars($article['category']); ?></span>
            <h3 class="article-title">
                <?php echo htmlspecialchars($article['title']); ?>
            </h3>
            <p class="article-excerpt">
                <?php echo htmlspecialchars($article['excerpt']); ?>
            </p>
            <div class="article-meta">
                <span class="author">By <?php echo htmlspecialchars($article['author']); ?></span>
                <span class="date"><?php echo $article['created_at']; ?> </span>
            </div>
            <a href="article.php?id=<?php echo $article['id']; ?>" class="read-more"> Read More → </a>
            <p class="view-count" style="color: #999; font-size: 0.9rem; margin-top: 10px;"> 👁 <span
class="views"> <?php echo $article['views']; ?> </span> views
            </p>
        </div>
    </article>
<?php
}
} else {
    echo "<p>No articles found.</p>";
}
// Close connection in the end of you file
// mysqli_close($conn);
?>
```

You should see all six articles displayed.

Comment out the article view counter in your JavaScript file so that you can see the article view loaded from database.

## 6. Article detail page

Display full article content: create 2 separated files

a. First file: article.php

- Create a copy of the techblog.php file, name as article.php

- In the beginning of the file, connect to get specific article from database

```

include 'db-connect.php';
// Get article ID from URL
$article_id = isset($_GET['id']) ? (int)$_GET['id'] : 0;
// Query to get specific article
$sql = "SELECT * FROM articles WHERE id = $article_id";
$result = mysqli_query($conn, $sql);
// Check if article exists
if (mysqli_num_rows($result) == 1) {
    $article = mysqli_fetch_assoc($result);
    // Increment view count
    $update_sql = "UPDATE articles SET views = views + 1 WHERE id = $article_id";
    mysqli_query($conn, $update_sql);
} else {
    echo "Article not found!";
    exit;
}

```

- Change the main content area (inside of main tag) to display an article content

```

<main class="article-container">
<article class="single-article">
    <!-- Article Header Image -->
    "
        class="article-header-image">
    <!-- Article Content -->
    <div class="article-body-content">
        <span class="article-category-badge">
            <?php echo htmlspecialchars($article['category']); ?>
        </span>
        <h1 class="article-main-title">
            <?php echo htmlspecialchars($article['title']); ?>
        </h1>
        <div class="article-metadata">
            <span>
                <strong>👤 Author:</strong>
                <span class="author-name">
                    <?php echo htmlspecialchars($article['author']); ?></span>
                </span>
            </div>
        </div>
    </div>
</article>
</main>

```

```

        <span> <strong><img alt="Calendar icon" data-bbox="382 103 401 118"/> Published:</strong>
<?php echo date('F d, Y', strtotime($article['created_at'])); ?>
    </span>
    <span><strong><img alt="Eye icon" data-bbox="308 151 327 166"/> Views:</strong>
    <?php echo number_format($article['views']); ?>
    </span>
</div>
<!-- Article Excerpt -->
<div style="background: #f8f9fa; padding: 20px; border-left: 4px solid #3498db; margin: 30px
0; font-style: italic;">
    <?php echo htmlspecialchars($article['excerpt']); ?>
</div>
<!-- Full Article Content -->
<div class="article-full-content">
<?php echo nl2br(htmlspecialchars($article['content'])); ?>
</div>
<!-- Back to Articles Button -->
<a href="techblog.php" class="back-to-articles">
    ← Back to All Articles</a>
</div>
</article>
</main>

```

- Close connection to database in the end  
 mysqli\_close(\$conn);

b. Second file: article.css

Step-by-step adding the style below and observe the results

.article-container

- max-width: 90%
- margin: 40px auto
- padding: 0 20px

.single-article

- background-color: #fff
- border-radius: 8px
- overflow: hidden
- box-shadow: 0 4px 6px rgba(0,0,0,0.1)

.article-header-image

- width: 100%
- height: 500px
- object-fit: cover

.article-body-content

- padding: 40px
- text-align: justify

.article-category-badge

- display: inline-block
- background-color: #3498db
- color: white

- padding: 8px 20px
  - border-radius: 20px
  - font-size: 0.9rem
  - margin-bottom: 20px
  - font-weight: 600
- .article-main-title
- font-size: 2.5rem
  - margin: 20px 0
  - color: #2c3e50
  - line-height: 1.3
- .article-metadata
- display: flex
  - gap: 25px
  - padding: 20px 0
  - border-bottom: 2px solid #ecf0f1
  - border-top: 2px solid #ecf0f1
  - margin: 20px 0
  - flex-wrap: wrap
- .article-full-content
- margin-top: 30px
  - line-height: 1.9
  - font-size: 1.1rem
  - color: #444
- .back-to-articles
- display: inline-block
  - margin-top: 40px
  - margin-right: 80px
  - padding: 12px 25px
  - background-color: #ecf0f1
  - color: #2c3e50
  - border-radius: 5px
  - font-weight: 600
  - transition: all 0.3s
- .back-to-articles:hover
- background-color: #3498db
  - color: white
  - transform: translateX(-5px)

Do not forget linking css file to your page

## 7. Adding category

a. Create a category table

- Run the query to create in phpMyAdmin

```
CREATE TABLE categories (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100) NOT NULL UNIQUE,  
  slug VARCHAR(100) NOT NULL UNIQUE,  
  description TEXT  
);
```

- Insert sample data into your table

```
INSERT INTO categories (name, slug, description) VALUES  
( 'Artificial Intelligence', 'artificial-intelligence', 'AI and machine learning topics'),  
( 'Web Development', 'web-development', 'Frontend and backend web technologies'),  
( 'Cybersecurity', 'cybersecurity', 'Security and privacy topics'),  
( 'Cloud Computing', 'cloud-computing', 'Cloud platforms and services'),  
( 'Mobile Development', 'mobile-development', 'iOS and Android development'),  
( 'Data Science', 'data-science', 'Data analysis and big data'),  
( 'Internet of Things', 'internet-of-things', 'IoT devices and smart technologies');
```

b. Update the sidebar in techblog.php to link to categories

```
<!-- In the sidebar categories widget -->  
<?php  
$cat_sql = "SELECT c.*,COUNT(a.id) as article_count FROM categories c  
  LEFT JOIN articles a ON c.name = a.category  
  GROUP BY c.id";  
$cat_result = $conn->query($cat_sql);  
?>  
<div class="widget categories">  
  <h3>Categories</h3>  
  <ul class="category-list">  
    <?php while($cat = $cat_result->fetch_assoc()): ?>  
      <li>  
        <a href="category.php?slug=<?php echo $cat['slug']; ?>">  
          <?php echo $cat['name']; ?> (<?php echo $cat['article_count']; ?>)  
        </a>  
      </li>  
    <?php endwhile; ?>  
  </ul>  
</div>
```

c. Display articles filtered by category

- Create a copy of the techblog.php file, name as category.php



- In the beginning of the file, connect to get specific category from database

```
require_once 'db-connect.php';
// Get category slug from URL
$category_slug = isset($_GET['slug']) ? $_GET['slug'] : '';
$category_slug = mysqli_real_escape_string($conn, $category_slug);
// Get category info
$cat_sql = "SELECT * FROM categories WHERE slug = '$category_slug'";
$cat_result = mysqli_query($conn, $cat_sql);
if (mysqli_num_rows($cat_result) == 0) {
    die("Category not found");
}
$category = mysqli_fetch_assoc($cat_result);
// Get articles in this category
$sql = "SELECT * FROM articles WHERE category = '{$category['name']}' ";
$result = mysqli_query($conn, $sql);
```

- Changing the main content area (inside of main tag)

```
<main class="category-container">
  <a href="techblog.php" class="back-button">← Back to Home</a>
  <div class="category-header">
    <h1 class="category-title"><?php echo htmlspecialchars($category['name']); ?></h1>
    <p class="category-description"><?php echo htmlspecialchars($category['description']); ?></p>
    <p class="category-count"><?php echo mysqli_num_rows($result); ?> article<?php echo
mysqli_num_rows($result) != 1 ? 's' : ''; ?></p>
  </div>
  <div class="articles-grid">
    <?php
    if (mysqli_num_rows($result) > 0) {
      while($article = mysqli_fetch_assoc($result)) { ?>
        <article class="article-card">
          " class="article-image">
          <div class="article-body">
            <span class="category"><?php echo htmlspecialchars($article['category']); ?></span>
            <h3 class="article-title">
              <?php echo htmlspecialchars($article['title']); ?>
            </h3>
            <p class="article-excerpt">
              <?php echo htmlspecialchars($article['excerpt']); ?>
            </p>
            <div class="article-meta">
              <span class="author">By <?php echo htmlspecialchars($article['author']); ?></span>
```

```

<span class="date"><?php echo $article['created_at']; ?> </span>
    </div>
    <a href="article.php?id=<?php echo $article['id']; ?>" class="read-more"> Read More →</a>
    <p class="view-count" style="color: #999; font-size: 0.9rem; margin-top: 10px;">  <span
class="views"> <?php echo $article['views']; ?> </span> views
    </p>
  </div>
</article>
<?php
}
} else {
    echo '<div class="empty-message">No articles in this category yet.</div>';
}
?>
</div>
</main>

```

d. Styling your page: [category.css](#)